



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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An Autonomous Institute, with NAAC "A" Grade
Department of Mechanical Engineering
Session 2025-26



VISION	MISSION
To be a center of excellence for learning and research in Mechanical Engineering.	1.To provide high quality, innovative and research environment in Mechanical Engineering. 2.To impart soft skill and hard skill to achieve institutional vision.

MID SEMETER EXAMINATION (MSE) - I

Course: B. Tech. in Mechanical Engineering

Subject: Product Design & Development

Subject Code:ME50003

Year/ Semester: - III/V

Max Marks: 20

Date: 11/08/2025

Duration: - 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Apply modern product design and development processes with understanding of design principles and validation.
2. Conduct customer needs analysis and implement market segmentation strategies effectively.
3. Generate and evaluate product concepts using structured methodologies and selection techniques.
4. Perform product analysis through benchmarking, reverse engineering, and competitive intelligence.
5. Implement sustainable design principles and manage product lifecycles comprehensively.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q.No.	Questions	Level/CO Mapped	Marks
Q.1	Name the curve used in technology forecasting to represent technology maturity.	L1/CO2	1
Q.2	State the primary difference between product development and product design.	L2/CO1	1
Q.3	Identify the three key approaches to achieving design efficiency.	L1/CO1	1
Q.4	List two essential tools used for gathering customer needs.	L3/CO2	1

SECTION B

1. All Questions are compulsory.
2. Answer in one/ two three sentences.

Q.No.	Questions	Level/CO Mapped	Marks
Q.5	Explain the role of standardization in product design and its benefits.	L2/CO1	2
Q.6	Describe the importance of customer involvement in product development	L3/CO2	2
Q.7	Distinguish between product testing and product validation .	L4/CO1	2
Q.8	Explain the significance of mission statement in customer-centric product development.	L3/CO2	2

SECTION C

1. Solve any two questions.

Q.No.	Questions	Level/CO Mapped	Marks
Q.9	Explain modern product development process and its differences from traditional approaches.	L4/CO1	4
Q.10	Analyze customer needs types and customer satisfaction models.	L4/CO2	4
Q.11	Explain essential factors for product design and role of standardization, simplification, and specialization.	L3/CO1	4